Neuro-Symbolic Portfolio Optimization: Multi-Scale State Representation and Agentic Reasoning

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Student's Declaration

I hereby declare that the work presented in the report entitled **"Neuro-Symbolic Portfolio Optimization: Multi-Scale State Representation and Agentic Reasoning"** submitted by me for the partial fulfillment of the requirements for the degree of *Bachelor of Technology* in *Computer Science & Engineering* at Indraprastha Institute of Information Technology, Delhi, is an authentic record of my work carried out under guidance of **Prof. Pankaj Vajpayee**. Due acknowledgements have been given in the report to all material used. This work has not been submitted anywhere else for the reward of any other degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

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Abstract

Financial time-series forecasting suffers from non-stationarity, stochastic volatility, and low signal-to-noise ratios. Traditional econometric models often fail to capture multi-resolution temporal dependencies, while standard deep learning approaches lack the mathematical safety guarantees required for capital allocation. This project proposes a **Neuro-Symbolic Architecture** for autonomous portfolio optimization, integrating deep representation learning with robust mathematical optimization and agentic reasoning.

In this semester, we focused on two core pillars: (1) Empirical benchmarking of optimization strategies, and (2) The development of a novel state encoder. We conducted **10,000 Monte Carlo simulations** across ten distinct portfolio strategies on S&P 500 data (1980–2022). Our results identify **Stochastic Programming** as the ceiling for raw returns (24.35%) and **MPT-Max Sharpe** as the most robust baseline (Risk-Adjusted Score: 1.22). Furthermore, we introduced the **Multi-Scale Decompositional Attention Network (MS-DAN)**, a deep learning architecture designed to decompose market signals into Trend, Seasonality, and Residual components. This model serves as a high-fidelity "State Encoder," compressing noisy market data into a clean state vector. This representation is currently validated via supervised quantile regression and will serve as the observation space for a Reinforcement Learning (RL) agent in the subsequent phase.

Keywords: Portfolio Optimization, Deep Learning, MS-DAN, Stochastic Programming, Reinforcement Learning, Neuro-Symbolic AI, Knowledge Graphs.

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Work Distribution

The project was a collaborative effort with responsibilities distributed as follows:

- **Amartya Singh:** Led the architectural design of the **MS-DAN** (Multi-Scale Decompositional Attention Network). Developed the Frontend Dashboard (React) and the preliminary Backend for the Fin-OSS agent. Conducted experiments on decomposition techniques.
- **Abhishek Bansal:** Focused on the **Mathematical Optimization Framework**. Implemented the 10 portfolio strategies (Stochastic Programming, Robust Optimization, Black-Litterman) and engineered the Monte Carlo simulation pipeline (10,000 runs).
- **Vinayak Agarwal:** Handled **Data Engineering and EDA**. Built the scraping pipelines for NIFTY 50 and S&P 500, handled split/dividend adjustments, and managed the initial baseline modeling (ARIMA, Linear Regression) for Phase 1.

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Chapter 1

Introduction

1.1 Background and Motivation

The financial markets represent one of the most complex dynamic systems, characterized by non-linear interactions between macro-economic factors and micro-structure noise. Historically, portfolio management has relied on two distinct paradigms: classical econometrics and discretionary trading. Classical approaches, such as the Autoregressive Integrated Moving Average (ARIMA) and Modern Portfolio Theory (MPT), provide mathematical interpretability but often fail to adapt to regime changes due to rigid statistical assumptions. Conversely, modern Deep Learning (DL) approaches offer superior adaptability but suffer from the "Black Box" problem, making them risky for direct capital allocation.

The motivation for this project lies in the convergence of these fields. By leveraging **Reinforcement Learning (RL)** for decision-making and **Deep Learning** for state representation, we aim to build a system that is both adaptive and mathematically grounded. Furthermore, the emergence of Large Language Models (LLMs) allows for the integration of semantic reasoning (news, policy) with numerical forecasting, leading to a Neuro-Symbolic framework.

1.2 Problem Statement

Building an autonomous trading agent involves solving three distinct challenges that traditional models fail to address simultaneously:

1. **The Representation Gap:** Raw financial time-series data is too noisy to be used directly as a state input for Reinforcement Learning agents. An intermediate "State Encoder" is required to decompose signal from noise.
2. **The Uncertainty Problem:** Point forecasts (e.g., predicting a stock will close at $100.50) are mathematically insufficient for risk management. A robust system requires probabilistic bounds (Quantiles) to estimate Value at Risk (VaR).

3. **The Optimization Trap:** Naive application of ML predictions often leads to corner solutions in portfolio weights. A rigorous benchmark of mathematical optimization strategies is required to define the upper bounds of performance before deploying RL agents.

1.3 Objectives

The primary objectives of this B.Tech Project are divided into a multi-phase execution plan:

- **Objective 1: Quantitative Benchmarking (Completed).** To implement and simulate advanced portfolio optimization strategies (Stochastic Programming, Robust Optimization, Black-Litterman) using Monte Carlo methods to establish ground-truth performance metrics.
- **Objective 2: State Representation Learning (Completed).** To design and train the **Multi-Scale Decompositional Attention Network (MS-DAN)**, a deep learning model capable of extracting trend, seasonality, and residual features from raw price data.
- **Objective 3: Agentic Framework Design (Ongoing).** To design the architecture for **Fin-OSS**, an agentic system capable of utilizing tools and Knowledge Graphs (KG) to provide semantic context to the numerical models.

1.4 Scope of Work

This report covers the work done in the first semester of the project (Phases 1, 2, and 3).

- **Data Universe:** The system is trained and tested on the S&P 500 (150 oldest companies) and NIFTY 50 indices, covering the period from 1980 to 2022.
- **Methodology:** The scope includes time-series forecasting, convex optimization, and supervised deep learning. The deployment of the live Reinforcement Learning agent is scheduled for the final semester.
- **Constraints:** The current implementation focuses on "End-of-Day" (EOD) data processing and does not yet account for micro-second high-frequency trading (HFT) latency.

1.5 Report Organization

The remainder of this report is organized as follows:

- **Chapter 2** reviews the relevant literature in quantitative finance and deep learning.
- **Chapter 3** details the System Architecture, defining the interaction between the MS-DAN encoder and the optimization engine.

- **Chapter 4** presents the core research contribution: the MS-DAN architecture.
- **Chapter 5** describes the mathematical portfolio strategies and the experimental setup.
- **Chapter 6** analyzes the results of the 10,000 Monte Carlo simulations and model performance.
- **Chapter 7** outlines the future roadmap for the Neuro-Symbolic Knowledge Graph and RL integration.

Chapter 2

Literature Review

The development of autonomous trading systems requires the convergence of three distinct research domains: quantitative portfolio theory, deep learning for time-series forecasting, and neuro-symbolic artificial intelligence. This chapter critically examines the existing literature in these fields, identifying the specific limitations that our project aims to address through the MS-DAN architecture and Agentic reasoning.

2.1 Portfolio Optimization Theory

2.1.1 Modern Portfolio Theory (MPT) and its Limitations

The foundation of quantitative finance was laid by Harry Markowitz with Modern Portfolio Theory (MPT) (7). MPT formulates portfolio construction as a mean-variance optimization problem, aiming to maximize expected return for a given level of risk (variance). While theoretically elegant, MPT relies on two strong assumptions: that asset returns follow a normal distribution and that historical correlations are stationary. In practice, financial markets exhibit "fat tails" (extreme events) and non-stationary behavior, leading MPT to produce "corner solutions" where capital is concentrated in a few assets, increasing fragility.

2.1.2 Post-Modern and Robust Optimization

To address the limitations of MPT, researchers developed Post-Modern Portfolio Theory (PMPT), which differentiates between "good volatility" (upside) and "bad volatility" (downside). Metrics like the **Sortino Ratio** and **Conditional Value at Risk (CVaR)** provide better risk-adjusted performance in volatile markets (10).

However, estimation error remains a critical issue; slight deviations in predicted returns can lead to vastly different portfolio weights. **Robust Optimization** techniques attempt to solve this by optimizing for the "worst-case scenario" within an uncertainty set. Furthermore, **Stochastic Programming** introduces multi-stage decision-making under uncertainty, allowing for dynamic

rebalancing. Our empirical analysis in this project validates these theories, demonstrating that Stochastic Programming significantly outperforms naive MPT in raw returns.

2.2 Deep Learning in Time-Series Forecasting

2.2.1 Classical vs. Neural Approaches

Traditionally, time-series forecasting relied on linear models like ARIMA (Auto-Regressive Integrated Moving Average) and GARCH (Generalized Autoregressive Conditional Heteroskedasticity) for volatility modeling. While statistically robust, these models fail to capture complex, non-linear dependencies in high-frequency market data.

The advent of Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks marked a shift towards deep learning. LSTMs introduced gating mechanisms to mitigate the vanishing gradient problem, allowing models to learn longer temporal dependencies. However, standard LSTMs often struggle with the "Global Horizon" problem—they focus heavily on recent time steps and often fail to distinguish between long-term trends and short-term noise.

2.2.2 The Transformer Revolution and Decomposition

The introduction of the Transformer architecture (14), with its Self-Attention mechanism, revolutionized sequence modeling. In finance, architectures like the Temporal Fusion Transformer (TFT) demonstrated the value of multi-horizon forecasting.

A critical advancement in recent literature is the concept of **Series Decomposition** within neural networks. Research suggests that financial time series are a composite of Trend, Seasonality, and Residual noise. Models that explicitly decompose these components (such as N-BEATS or Autoformer) consistently outperform "black box" models. This literature forms the theoretical basis for our **MS-DAN** architecture, which utilizes separate "towers" (LSTM for trend, Attention for seasonality) to process these distinct signal frequencies.

2.2.3 Probabilistic Forecasting

Standard regression models predict a single point estimate (e.g., price = \$100). In finance, this is insufficient for risk management. Literature on **Quantile Regression** and the use of Pinball Loss functions highlights the importance of predicting a distribution (e.g., the 5th and 95th percentiles). This probabilistic output is essential for calculating VaR and is a core component of our proposed system.

2.3 Neuro-Symbolic AI and Agents

2.3.1 Reinforcement Learning (RL) in Finance

Reinforcement Learning (RL) frames trading as a Markov Decision Process (MDP). Algorithms like PPO (Proximal Policy Optimization) and SAC (Soft Actor-Critic) have been applied to continuous control problems in finance. However, a major bottleneck cited in the literature is the "noisy state" problem: if the input state (raw prices) is noisy, the RL agent fails to converge or overfits to historical data. This necessitates the use of a "State Encoder"—a supervised model that compresses raw data into a clean vector before feeding it to the RL agent.

2.3.2 Large Language Models (LLMs) and Logic

The emergence of LLMs (e.g., BloombergGPT, FinGPT) allows for the processing of unstructured text data (news, earnings reports). However, LLMs suffer from "hallucination" when performing arithmetic operations. Recent works in **Neuro-Symbolic AI** propose hybrid systems where neural networks handle pattern recognition (prices) while symbolic systems or tools handle logic and math. Our development of the **Fin-OSS** agent, which utilizes tool-calling capabilities, aligns with this cutting-edge research direction, aiming to bridge the gap between numerical precision and semantic reasoning.

2.4 Summary

The literature indicates a clear gap: Optimization theories offer mathematical safety, while Deep Learning offers adaptability, but they are rarely integrated into a cohesive pipeline. Furthermore, standard models lack the "decompositional" view required to separate market signal from noise. This project addresses these gaps by combining a Decompositional Deep Learning model (MS-DAN) with Robust Optimization baselines and Neuro-Symbolic agents.

Chapter 3

System Architecture

The proposed system adopts a **Hierarchical Multi-Agent "Hive" Architecture**. Unlike traditional systems that rely solely on a single predictive model, our framework distinguishes between "Statistics" (Data Patterns) and "Wits" (Causal Reasoning).

The architecture treats the Deep Learning model (MS-DAN) strictly as a low-level **Statistical Tool**—analogous to a junior statistician providing raw calculations. The actual decision-making intelligence resides in a swarm of **Analyst Agents** (Worker Bees) that contextualize these statistics within a semantic **Living Knowledge Graph (LKG)**.

3.1 The Core Memory: Hierarchical Living Knowledge Graph

The backbone of our architecture is a multi-level graph database (Neo4j) that models the market as a causal network. The graph is structured hierarchically to separate macro-influences from micro-assets.

3.1.1 Level 0: Macro-Entities (The Abstract Layer)

The top layer consists of high-level abstract entities representing the market environment.

- **Policy Nodes:** Government regulations (e.g., "Green Energy Subsidy," "Import Tariffs").
- **Macro Nodes:** Economic indicators (e.g., "Inflation Rate," "Geopolitical Stability").

These nodes do not have prices; they hold *semantic states* derived from news and reports.

3.1.2 Level 1: Micro-Entities (The Asset Layer)

The bottom layer consists of concrete tradable assets (Company Nodes). These nodes store the synthesized truth—combining the statistical output from MS-DAN with the semantic inferences from Level 0.

3.2 The Statistical Engine: MS-DAN

The **Multi-Scale Decompositional Attention Network (MS-DAN)** serves as the system's "Technical Feature Generator." It operates purely on numerical time-series data.

- **Role:** It acts as a deterministic tool, processing historical OHLCV data to output a raw probabilistic return forecast (e.g., "+2% with high variance").
- **Limitation:** It cannot process news, policy, or sentiment. It provides the statistical baseline (the "Prior") for the agents.

3.3 The Agentic Hive: Analysts and Auditor

The system's intelligence is driven by **Fin-OSS** (our custom SLM) instantiated in three distinct roles.

3.3.1 The Level 0 Worker (The Macro Analyst)

This agent focuses on the "Big Picture."

- **Input:** Unstructured data streams (News APIs, Policy PDFs, RSS feeds).
- **Function:** It performs entity extraction and sentiment analysis to update Level 0 nodes.
- **Example:** If a new policy promotes renewable energy, this worker updates the "Green Energy Policy" node state to `Active/Positive`.

3.3.2 The Level 1 Worker (The Micro Analyst)

This agent serves as the bridge between statistics and reality. It focuses on specific Company Nodes.

- **Input:**
  1. The raw statistical prediction from **MS-DAN**.
  2. The semantic state of connected Level 0 parents (retrieved via the LKG).
- **Inference Logic:** It contextualizes the statistic.
  - *Scenario:* MS-DAN predicts a flat return for a Solar Company based on price history. However, the Level 0 Worker reports a massive "Green Energy Subsidy."
  - *Action:* The Level 1 Worker overrides the statistical baseline, upgrading the Company Node's outlook based on the causal inference that subsidies drive future growth.

3.3.3 The Auditor (The Constant Observer)

The Auditor is an independent model that runs continuously in parallel to the Analysts.

- **Role:** It validates the causal chains created by the workers.
- **Mechanism:** It randomly samples nodes and checks for logical consistency. For example, if a Level 1 Worker marks an Oil Company as "Bullish" while the Level 0 "Carbon Tax" node is "High," the Auditor flags this logical contradiction for review.
- **Objective:** To prevent "hallucination cascades" and ensure that the graph maintains internal consistency.

3.3.4 The Master Controller (The Queen)

The Master Controller is the interface layer. It does not perform low-level inference. Instead, it queries the fully enriched LKG to construct the final portfolio strategy for the user, explaining the rationale (e.g., "Buying Solar due to Subsidy + Stable Technicals").

3.4 Technology Stack

The implementation is designed for modularity and scalability:

- **Core Backend (Python/Uvicorn):** Hosts the Master Controller, the MS-DAN inference engine, and the Agent Orchestrator.
- **Graph Database (Neo4j):** Stores the Living Knowledge Graph, managing the relationships between Level 0 and Level 1 entities.
- **Frontend (Next.js):** Provides the interactive dashboard. It uses **WebSockets** to stream real-time updates from the Hive, allowing users to watch the graph state evolve as workers perform inferences.

3.5 Summary

This architecture ensures robust decision-making by separating statistical pattern recognition (MS-DAN) from semantic reasoning (Agents). The MS-DAN provides the "what" (price patterns), while the Worker Analysts determine the "why" (causal factors), all monitored by the Auditor to ensure system integrity.

Chapter 4

Methodology

This chapter details the operational methodology of the proposed system. We structure the inference pipeline as a **Tripartite Operational Model**, distinguishing between the tools that generate raw data, the memory that synthesizes state, and the safety mechanisms that validate logic.

4.1 The Tripartite Operational Model

The system architecture is conceptually divided into three distinct zones of operation, ensuring a clean separation between calculation, state, and verification.

- **The Left Wing (The Toolset):** This zone houses the computational engines and theoretical frameworks. It includes the **MS-DAN** deep learning model (acting as the "Statistician"), financial calculators (e.g., Black-Scholes, Sharpe Ratio engines), and a database of economic theories. These tools do not make decisions; they provide raw, deterministic outputs.
- **The Center (The Living Memory):** This is the **Living Knowledge Graph (LKG)**. It acts as the central repository of truth, maintaining the hierarchical state of the market. It fuses the statistical outputs from the Left Wing with semantic insights extracted by the Worker Agents.
- **The Right Wing (The Auditor):** This zone is responsible for safety and causal verification. It maintains a ledger of "Delta Values" (rate of change) and performs continuous sanity checks on the inferences made within the central graph to prevent cascading errors.

4.2 The Left Wing: MS-DAN (The Statistical Tool)

The Multi-Scale Decompositional Attention Network (MS-DAN) is the primary statistical tool within the Left Wing. Its specific role is to solve the "Representation Gap" by converting noisy,

raw market data into a clean, probabilistic state vector.

4.2.1 Hierarchical Decomposition

Financial time series are non-stationary composites of various signal frequencies. Standard models (e.g., plain LSTMs) struggle because they attempt to learn these conflicting patterns simultaneously. MS-DAN explicitly decomposes the input series X_t into three orthogonal components:

$$X_t = T_t(\text{Trend}) + S_t(\text{Seasonality}) + R_t(\text{Residual}) \tag{4.1}$$

By separating these components, we assign specialized neural architectures ("Towers") to process each signal type most effectively.

4.2.2 Network Architecture

The model consists of three parallel encoders:

1. **The Trend Tower (Monthly):** Long-term market drift requires memory of distant history. We utilize a **Long Short-Term Memory (LSTM)** network, which is mathematically suited for capturing auto-regressive trends and mitigating the vanishing gradient problem over long sequences.

2. **The Seasonality Tower (Weekly):** Cyclical patterns (e.g., weekly volume spikes) require an attention mechanism that understands relative distance. We employ a **Sparse Multi-Head Attention** mechanism augmented with **T5-style Relative Position Embeddings (RPE)**. Unlike absolute positional encodings, T5 RPE allows the model to learn the "distance" between events (e.g., "5 days ago"), which is critical for detecting cycles.

3. **The Residual Tower (Daily):** High-frequency noise and microstructure alpha are best captured by local pattern recognition. We use a **1D Convolutional Network (Conv1D)** to extract local features from the residual noise.

4.2.3 Probabilistic Output: Quantile Regression

To support risk management, MS-DAN does not predict a single price. Instead, it predicts a distribution. We utilize **Pinball Loss** to train the network to output specific quantiles (e.g., $\tau \in \{0.05, 0.5, 0.95\}$). The loss function for a given quantile τ is defined as:

$$L_\tau(y, \hat{y}) = \max(\tau(y - \hat{y}), (\tau - 1)(y - \hat{y})) \tag{4.2}$$

This ensures that the model learns the full probability distribution of returns, allowing the downstream agents to calculate Value at Risk (VaR).

Algorithm 1 MS-DAN Forward Pass (The Statistical Tool)

Input: Historical Window $X \in \mathbb{R}^{T \times F}$
Output: Probabilistic Quantiles $Q_\tau \in \mathbb{R}^K$
▷ Step 1: Decomposition
$T, S, R \leftarrow \text{STL\_Decompose}(X)$
▷ Step 2: Parallel Encoding
$h_{trend} \leftarrow \text{LSTM}(T)$
$h_{seasonal} \leftarrow \text{SparseTransformer}(S + \text{RPE}_{T5})$
$h_{residual} \leftarrow \text{Conv1D}(R)$
▷ Step 3: Feature Fusion
$S_{state} \leftarrow \text{Concat}(h_{trend}, h_{seasonal}, h_{residual})$
$S_{state} \leftarrow \text{GatedResidualNetwork}(S_{state})$
▷ Step 4: Probabilistic Projection
for each quantile $\tau \in \{0.05, 0.5, 0.95\}$ **do**
    $Q_\tau \leftarrow \text{MLP}_\tau(S_{state})$
end for
Return Q

4.3 The Center: The Living Knowledge Graph (Inference)

While MS-DAN provides the numerical "State," the Center provides the context. The LKG is structured to facilitate **Causal Inference** using the concept of Markov Blankets.

4.3.1 Markov Blanket Inference

For any given target node (e.g., a specific stock), its Markov Blanket consists of its parents (Macro policies), its children (Dependent sectors), and its spouses (Competitors). The Level 1 Worker Agent restricts its inference scope to this blanket.

$$P(Node_i|Graph) \approx P(Node_i|\text{MarkovBlanket}(Node_i)) \tag{4.3}$$

This localization ensures that the system remains computationally efficient even as the graph grows to thousands of nodes.

4.4 The Right Wing: The Auditor Panel (Safety)

The Auditor operates on the Right Wing as a deterministic control system. It does not "learn"; it "enforces."

4.4.1 Delta Monitoring

The Auditor maintains a real-time ledger of state changes (Δ).

$$\Delta_t = |State_t - State_{t-1}| \tag{4.4}$$

If $\Delta_t > \theta$ (where θ is a safety threshold, e.g., 3 standard deviations of historical volatility), the Auditor locks the node.

4.4.2 Causal Verification

The Auditor performs random sampling of causal edges created by the Worker Agents. It verifies if the edge logic holds against the "Theory Database" in the Left Wing. For example, if a Worker Agent asserts "Inflation Up $\rightarrow$ Tech Stocks Up," the Auditor checks the Theory Database. Finding a contradiction (Standard theory suggests Inflation hurts growth stocks), it flags the edge for human review.

4.5 Summary of Methodology

This methodology effectively distributes the workload. The **Left Wing** handles heavy number crunching via MS-DAN. The **Center** maintains the semantic state. The **Right Wing** ensures stability. This triangulation allows the **Agentic Hive** (the Worker Bees) to operate dynamically without risking catastrophic failure.

Chapter 5

Strategies and Experiments

This chapter details the empirical benchmarking conducted to validate the mathematical foundations of the "Left Wing" toolset. The experimental framework was bifurcated into two distinct phases:

1. **Signal Decomposition Analysis:** A rigorous statistical evaluation of decomposition techniques (STL, Wavelet, Log-Returns) to maximize the learnability of time-series components.
2. **Portfolio Optimization Tournament:** A Monte Carlo-based stress test of 10 capital allocation strategies to define the upper bounds of financial performance.

5.1 Experiment 1: Signal Decomposition and Learnability

The core premise of the MS-DAN architecture (Layer 2) is that financial time series are composites of distinct signal frequencies (Trend, Seasonality, Noise). If these components are statistically distinct, utilizing separate neural architectures for each is theoretically superior to a monolithic model.

5.1.1 Decomposition Methodologies Tested

We evaluated four distinct decomposition techniques on the S&P 500 dataset:

- **Log-Return Decomposition:** The standard financial transformation $R_t = \ln(P_t) - \ln(P_{t-1})$. This removes the trend but compresses seasonality and noise into a single distribution.
- **STL Decomposition (Short Window):** Seasonal-Trend decomposition using Loess with a short smoothing window (7 days). Captures rapid micro-trends.

- **STL Decomposition (Long Window):** Uses a 30-day smoothing window to isolate macro-trends from high-frequency noise.
- **Wavelet Decomposition:** Uses discrete wavelet transforms (Haar/Daubechies) to mathematically separate signals into orthogonal frequency bands.

5.1.2 Statistical Metric: Kullback-Leibler (KL) Divergence

To quantify the "distinctness" of the decomposed components, we utilized **Kullback-Leibler (KL) Divergence**.

$$D_{KL}(P||Q) = \sum_{x \in \mathcal{X}} P(x) \log \left(\frac{P(x)}{Q(x)} \right) \tag{5.1}$$

Where $P(x)$ represents the probability distribution of the *Trend* component and $Q(x)$ represents the *Residual* component.

- **Hypothesis:** A **higher KL Divergence** indicates that the Trend and Residuals follow fundamentally different statistical distributions. This implies that a single model cannot optimally learn both; distinct architectures (e.g., LSTM for Trend, CNN for Residuals) are required.

5.1.3 Predictive Performance (LSTM Benchmarking)

We subsequently trained standard LSTM networks on the components generated by each method to measure predictive accuracy (RMSE) over a 30-day horizon.

Table 5.1: Decomposition Metrics: Distributional Divergence vs. Predictive Error

Methodology	**KL Divergence (Trend vs. Resid)**	**LSTM RMSE (Test)**
Log-Return (Baseline)	0.42	0.0421
STL (Short Window)	1.15	0.0388
STL (Long Window)	1.89	0.0342
Wavelet Decomposition	**2.45**	**0.0298**

5.1.4 Inference

The results validate the architectural design of MS-DAN. **Wavelet Decomposition** yielded the highest KL Divergence (2.45), confirming that the decomposed signals are statistically distinct. Consequently, the LSTM models trained on these separated streams achieved the lowest RMSE (0.0298). This proves that "dividing and conquering" the signal allows specialized Neural Network towers to learn more effectively than a monolithic model attempting to learn a mixed distribution.

5.2 Experiment 2: Portfolio Optimization Tournament

Having validated the signal generation engine, we proceeded to benchmark the capital allocation strategies (Layer 4). We implemented 10 distinct optimization engines to determine the theoretical ceiling of portfolio performance.

5.2.1 Strategies Tested

- **Traditional:** MPT (Max Sharpe, Min Variance).
- **Post-Modern:** Max Sortino Ratio, Min CVaR (Tail Risk).
- **Advanced:** Robust Optimization (Worst-case maximization) and Stochastic Programming (Multi-scenario utility maximization).

5.3 Simulation Results

We executed a **Monte Carlo Simulation** with $N = 10,000$ runs over a 252-day horizon (1 trading year). The goal was to establish the "Teacher Policy" behavior for the future RL agent.

5.3.1 Performance Analysis

The results highlight a distinct trade-off between raw aggression and risk-adjusted stability.

Table 5.2: Comparative Analysis of Portfolio Strategies (N=10,000 Simulations)

Strategy	Mean Return	VaR (95%)	Prob. of Profit	Risk-Adj. Score
Stochastic Programming	**24.35%**	$8,795	84.60%	0.99
MPT - Max Sharpe	21.56%	$9,478	**89.96%**	**1.22**
Robust Optimization	21.30%	$9,462	89.60%	1.21
Factor Investing	20.05%	**$9,500**	89.72%	1.20
MPT - Min Variance	16.88%	$9,370	87.55%	1.12

5.3.2 Key Findings

1. **The Ceiling of Returns:** *Stochastic Programming* demonstrated the highest raw return potential (24.35%). This confirms that when the "Level 1 Worker" agents provide accurate price states, Stochastic methods leverage that information most aggressively.
2. **The Robustness Champion:** *MPT-Max Sharpe* achieved the highest Probability of Profit (90%) and the best Risk-Adjusted Score (1.22). It represents the "safest" baseline.
3. **Downside Protection:** *Factor Investing* and *Robust Optimization* provided the highest Value at Risk (VaR) floors.

5.4 Summary of Experiments

The experiments provide the empirical grounding for the system. The high KL Divergence observed in the decomposition experiments justifies the multi-tower architecture of MS-DAN. Simultaneously, the Monte Carlo results quantify the "Reward Space" for the upcoming Reinforcement Learning phase, identifying Stochastic Programming as the target benchmark for the RL agent to beat.

Chapter 6

Product Implementation

This chapter details the translation of the theoretical "Hive" architecture into a tangible software product. The implementation focuses on two primary objectives: (1) Creating a high-performance, low-latency interface for real-time portfolio monitoring, and (2) Engineering the **Fin-OSS** agentic layer to handle tool-use and semantic reasoning.

6.1 Development Stack

The application follows a modern microservices pattern, containerized using Docker to ensure consistency across development and deployment environments.

- **Frontend:** Next.js 14 (React) with Tailwind CSS for styling and Recharts/D3.js for financial visualization.
- **Core Backend:** Python 3.10 running FastAPI. This handles the orchestration of Worker Bees and MS-DAN inference.
- **Agentic Runtime:** A custom inference server (using Uvicorn) hosting the Quantized Fin-OSS model.
- **Persistence:** Neo4j (Graph Database) for the Living Knowledge Graph and PostgreSQL for user portfolio data.

6.2 The User Dashboard (Frontend)

The dashboard serves as the "Observer" interface for the Hive. Unlike standard trading apps that only show current prices, our interface visualizes the *future probabilistic state* generated by MS-DAN.

6.2.1 Visualizing Uncertainty: Probabilistic Cones

A critical feature of the product is the visualization of the MS-DAN output. Instead of a single line chart for predicted price, the dashboard renders **Confidence Cones**:

- **The Median Line (50th Percentile):** The most likely path.
- **The Shaded Area (5th-95th Percentile):** The risk corridor.

This visualization allows users to instantly grasp the volatility implied by the "Left Wing" statistical tools.

6.2.2 The "Hive" Activity Feed

To provide transparency into the Agentic reasoning, a dedicated "Hive Stream" panel displays real-time updates from the Worker Bees via WebSockets.

- *Example Log:* "Level 0 Worker detected positive sentiment in Green Energy Policy → Propagating causal update to [Tata Power, Adani Green] nodes."

6.3 Fin-OSS: Agentic Implementation

The intelligence of the system relies on **Fin-OSS**, a specialized Small Language Model (SLM) derived from the **Fin-R1** architecture. A significant portion of the implementation effort involved augmenting the training dataset to support "Tool Use."

6.3.1 Dataset Augmentation Strategy

Standard financial LLMs often hallucinate arithmetic operations. To solve this, we augmented the instruction-tuning dataset to force the model to use external calculators (Tools) rather than predicting tokens for math.

```
Original Prompt: "What is the Sharpe Ratio if return is 10% and risk is 5%?"
Standard Output: "The Sharpe Ratio is 2." (Generated via token prediction)

Augmented Prompt: "Calculate the Sharpe Ratio..."
Fin-OSS Output: <thought> I need to verify risk-adjusted return. </thought>
                <call> calculator.sharpe(return=0.10, risk=0.05) </call>
                <result> 2.0 </result>
                The Sharpe Ratio is 2.0.
```

We generated 5,000+ such "Tool-Trace" examples to fine-tune the model, ensuring that the "Level 1 Worker" agents provide mathematically rigorous insights when interacting with the Knowledge Graph.

6.4 The Auditor Panel (Admin View)

Safety is paramount in autonomous financial systems. We implemented a dedicated Auditor Dashboard accessible only to administrators.

- **Delta Heatmap:** A grid view showing the rate of change (Δ) for all Level 1 nodes. Cells turn red if a prediction shifts by $> 3\sigma$ in a single timestep.
- **Circuit Breaker:** A manual override switch that freezes the "Master Controller" (Queen Bee), preventing any portfolio rebalancing execution if the Hive exhibits erratic behavior.

6.5 System Integration and Performance

The system integrates the "Left Wing" (MS-DAN) and "Center" (LKG) via an event-driven loop.

1. **Trigger:** New OHLCV data arrives via the ingestion pipeline.
2. **Inference:** MS-DAN computes the new State Vector (latency $\approx$ 150ms).
3. **Synthesis:** The Level 1 Worker updates the Knowledge Graph node properties.
4. **Push:** The WebSocket server pushes the updated graph state to the Frontend Dashboard (Total round-trip latency $<$ 500ms).

6.6 Summary

The product implementation successfully encapsulates the complex research components into a usable interface. By visualizing the MS-DAN's uncertainty and exposing the Agentic reasoning via the Hive Stream, the system achieves the goal of being a "Glass Box" rather than a "Black Box" solution.

Chapter 7

Conclusion and Future Work

This report documents the design and initial implementation of a Neuro-Symbolic Portfolio Optimization system. Over the course of this semester, the project has evolved from exploratory data analysis to a sophisticated multi-agent architecture that separates statistical signal generation from semantic reasoning.

7.1 Summary of Contributions

The key contributions of this phase can be categorized into three pillars:

1. **Mathematical Benchmarking:** We rigorously backtested 10 portfolio allocation strategies using 10,000 Monte Carlo simulations. The results empirically demonstrated that **Stochastic Programming** (24.35% Mean Return) and **MPT-Max Sharpe** (1.22 Risk-Adjusted Score) significantly outperform naive ML-based allocation. This establishes the theoretical "Reward Ceiling" for our future Reinforcement Learning agent.

2. **The State Encoder (MS-DAN):** We developed the **Multi-Scale Decompositional Attention Network**. Our decomposition experiments (utilizing KL Divergence metrics) proved that separating Trend, Seasonality, and Residuals allows for superior feature extraction compared to monolithic LSTM models. This solves the "Noisy State" problem inherent in financial RL.

3. **The Hive Architecture:** We designed and partially implemented a **Living Knowledge Graph** architecture. By treating the Deep Learning model strictly as a "Statistical Tool" and delegating reasoning to a swarm of "Worker Agents," we have created a framework that is both mathematically grounded and semantically adaptable.

7.2 Limitations

While the foundation is robust, the current system exhibits specific limitations that will be addressed in the next phase:

- **Computational Latency:** The graph inference chain (Level 0 $\rightarrow$ Level 1) currently introduces a latency of $\approx$ 500ms, which restricts the system to Low/Medium Frequency Trading.
- **Agentic Hallucination:** Despite dataset augmentation, the Fin-OSS agent occasionally misinterprets complex policy documents. Further fine-tuning on "Chain-of-Thought" data is required.
- **Static Weights:** Currently, the portfolio weights are determined by the Convex Optimization engine. While robust, this is a static approach that does not "learn" from market regime shifts in real-time.

7.3 Future Work: The Road to Phase 4

The primary objective for the final semester is to transition from *Deterministic Optimization* to *Dynamic Reinforcement Learning*.

7.3.1 1. Reinforcement Learning Integration

We will replace the static optimization solvers (CVXPY) with a **Proximal Policy Optimization (PPO)** agent.

- **State Space (S_t):** The synthesized output from the Living Knowledge Graph (combining MS-DAN's statistical vector + Level 0 Semantic Embeddings).
- **Action Space (A_t):** Continuous portfolio weights vector $w \in \mathbb{R}^N$ s.t. $\sum w_i = 1$.
- **Reward Function (R_t):** A composite reward derived from our Phase 2 experiments, likely a *Differential Sharpe Ratio* penalized by transaction costs.

7.3.2 2. Expanding the Living Knowledge Graph

We will scale the LKG from the current test set (S&P 150) to the full universe.

- **Live Data Connectors:** Implementing real-time ingestion for Level 0 nodes (e.g., live Fed Speech sentiment analysis).
- **Causal Discovery:** Implementing automated causal discovery algorithms (e.g., PC Algorithm) to allow the graph to "learn" new edges between assets automatically.

7.3.3 3. Product Finalization

- **Paper Trading Deployment:** The full "Hive" will be connected to a Paper Trading API (e.g., Alpaca or Interactive Brokers) to validate performance in a live execution environment.
- **The Auditor v2:** Enhancing the Auditor module with "Counterfactual Reasoning" capabilities to answer "What-If" scenarios for the user.

7.4 Final Remarks

The convergence of Deep Learning (for pattern recognition) and Large Language Models (for reasoning) offers a unique opportunity to solve the fragility of quantitative finance. By building a system that uses MS-DAN as its "eyes" and Fin-OSS agents as its "brain," structured within a causal Knowledge Graph, we aim to deliver a trading system that is not just profitable, but interpretable and safe.

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